



CLOUD ADOPTION PLAYBOOK

ACCELERATOR PLATFORM WITH TURNKEY SOLUTIONS

*ACCELERATING THE ENTERPRISE CLOUD ADOPTION AND DIGITAL MODERNIZATION JOURNEY
SECURELY AND EFFICIENTLY*

VINAY MANNE, CHIEF TECHNOLOGY OFFICER

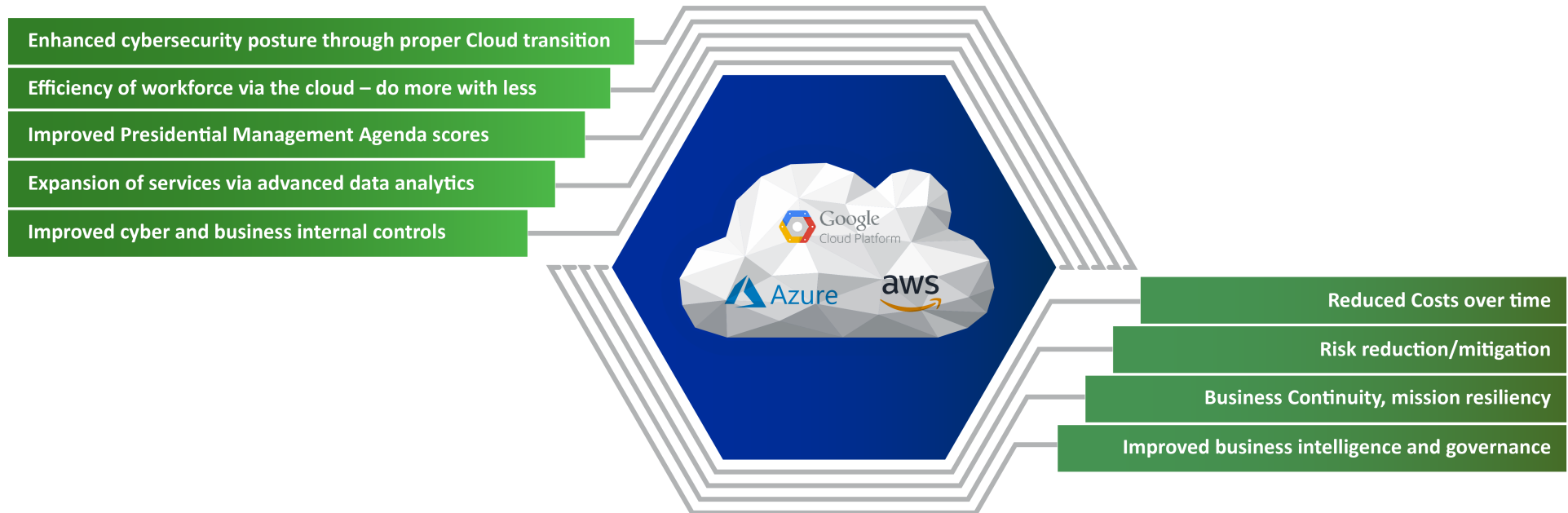
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WHY CLOUD ?



WHY CLOUD ?

CAPABILITY- CLOUD ENABLES USE OF MODERN TECHNOLOGIES

- **Capability** is the ability to leverage all the new service delivery and decision support capabilities of modern computing including “big Data”, analytics, IoT, and AI/ML using 5G to support your mission. If your apps and information are not cloud-enabled, these capabilities will not be available.

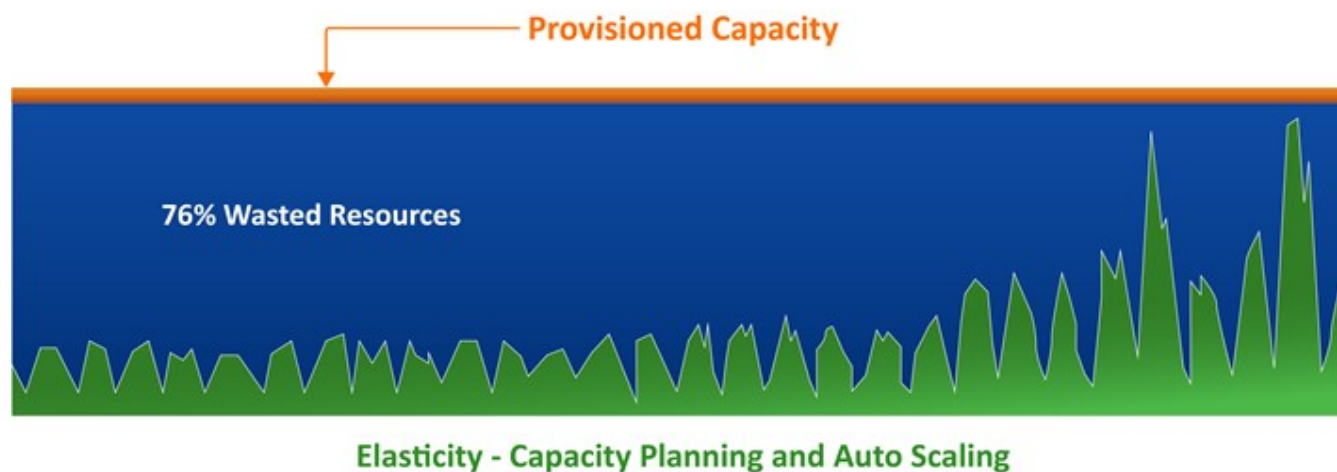


While your current technology may seem OK to you, what does it look like to the new generation of workers and customers who do everything on their mobile phone?

WHY CLOUD ?

CAPACITY - ON DEMAND RESOURCES CONSUMPTION

- **Capacity** refers to the fact the current data center architecture requires you to “guess” what your level of user demand on your systems will be and provision for the highest level. Cloud allows you the flexibility to get support at the level of your “real time” needs, in order to provide the best user experience.

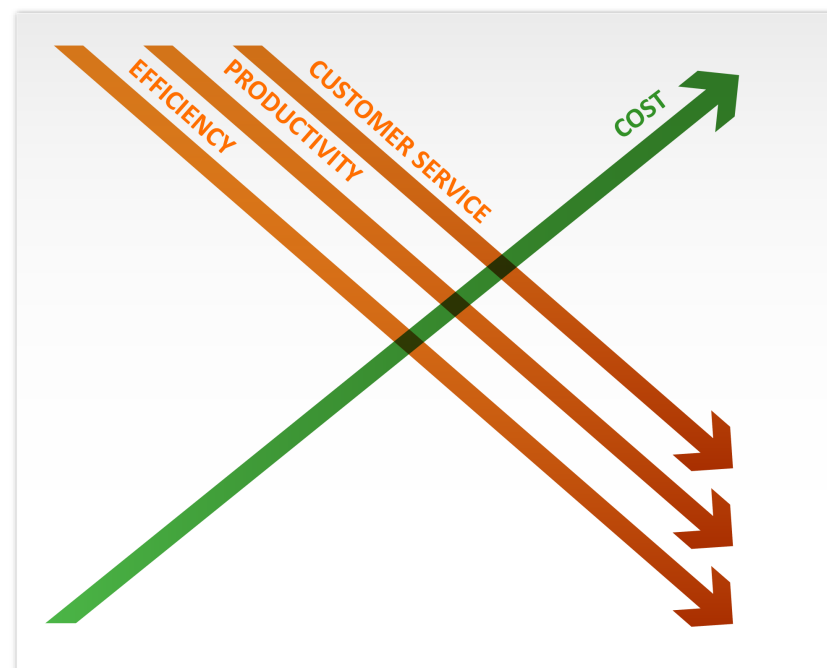


Pay for what you need, while having confidence your systems will be always be accessible for customers

WHY CLOUD ?

RISK - LEGACY HOSTING POSES INCREASING STRATEGIC RISK

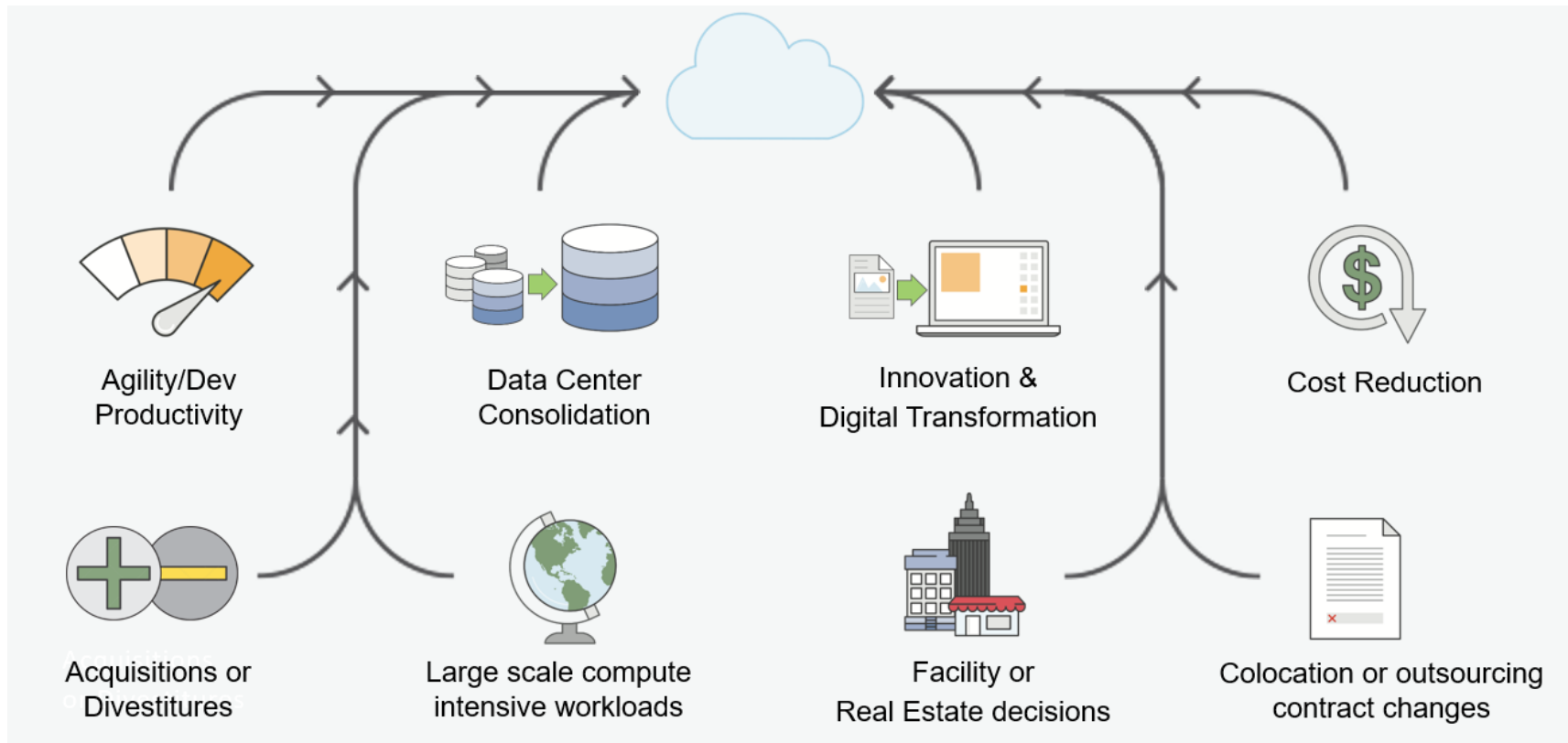
- **Risk** is that you will be stuck on old technology that no one supports anymore, and maintenance costs will continue to rise.
- Worse yet, these old systems will become security risks.
- No Access to Benefits of the cloud
- Limited Choices



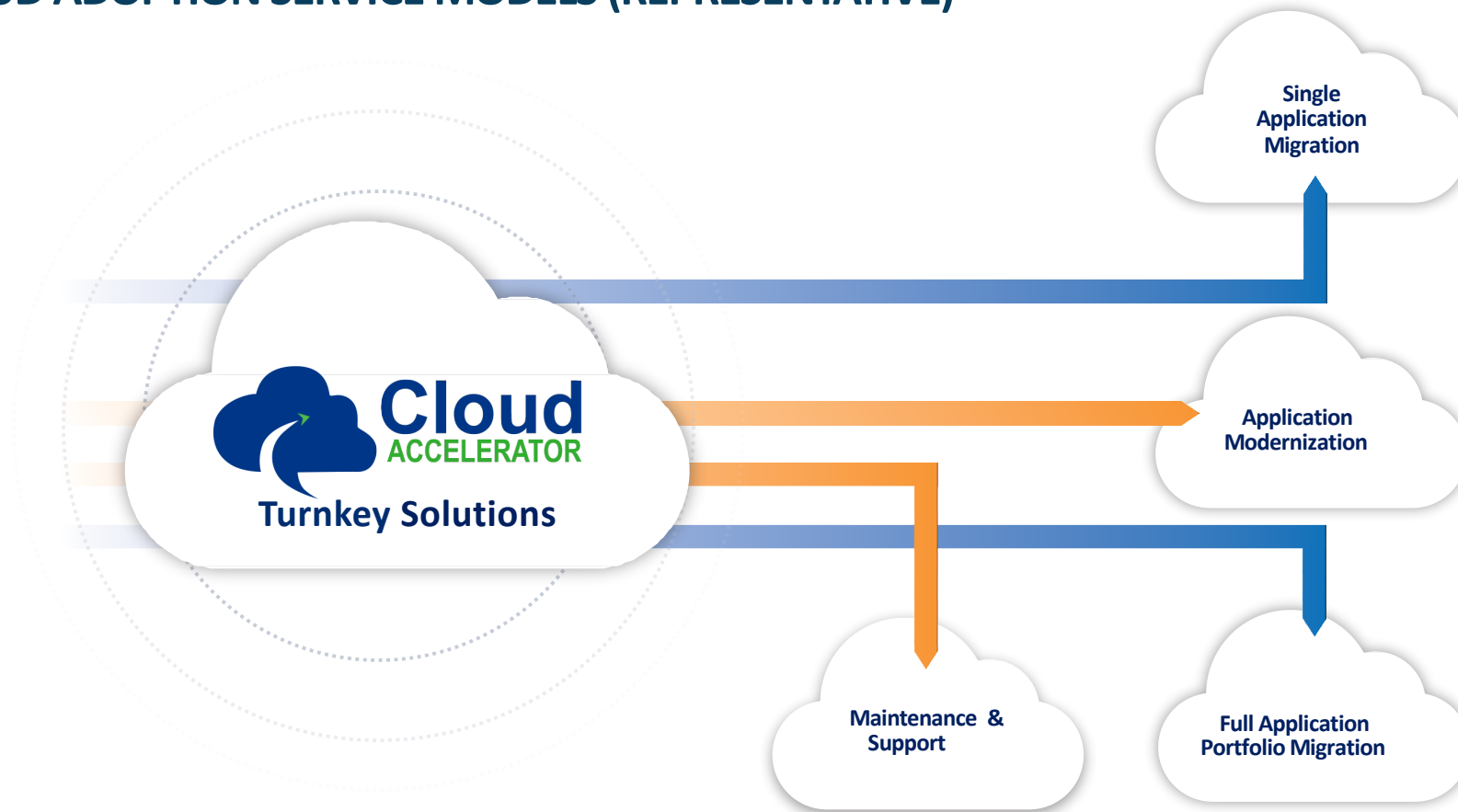
In the end, you will have to make a dramatic and more costly decision, while maintaining inefficient old systems.

WHY CLOUD ?

CLOUD ADOPTION – COMMON DRIVERS SUMMARIZED



CLOUD ADOPTION SERVICE MODELS (REPRESENTATIVE)



CLOUD ADOPTION FEATURES, VALUE-ADD AND OUTCOMES

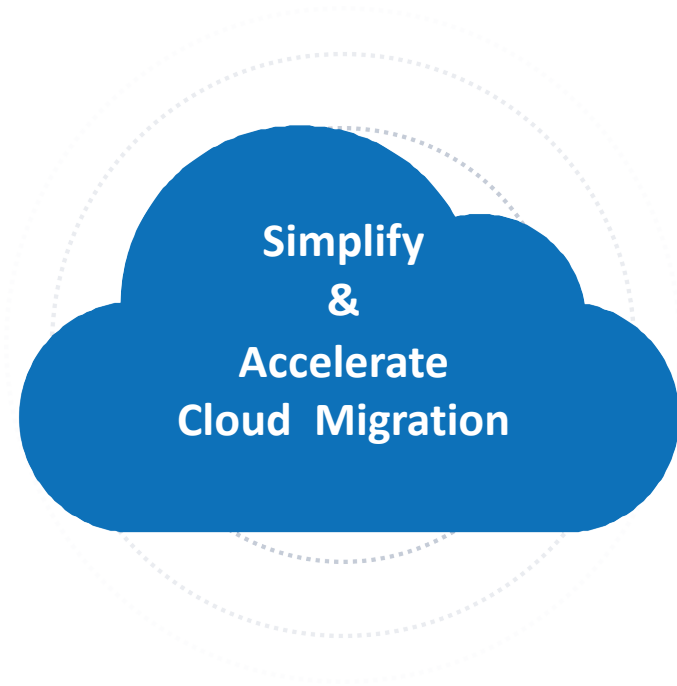
Features

Managed Cloud
Services

Turn-Key
Migration

DevSecOps
Automation

Flexible Pricing
Models



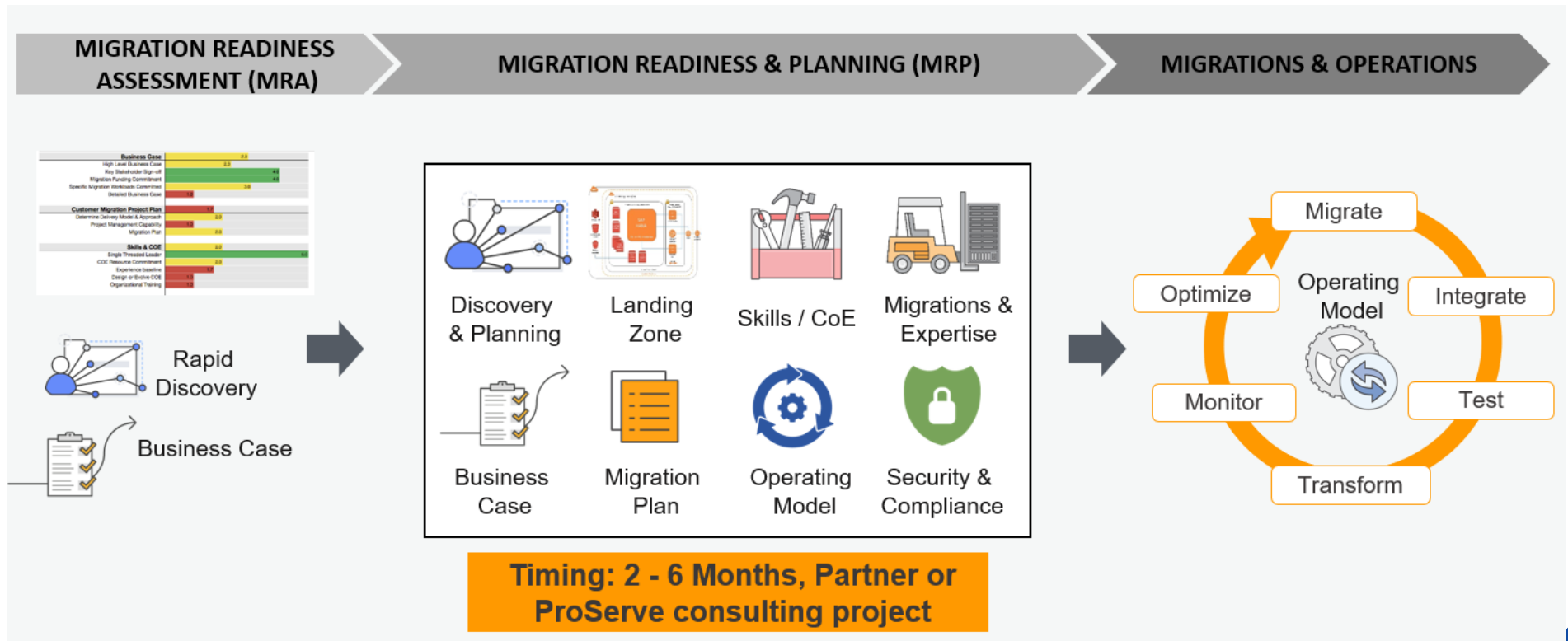
**Simplify
&
Accelerate
Cloud Migration**

Value Add and Outcomes

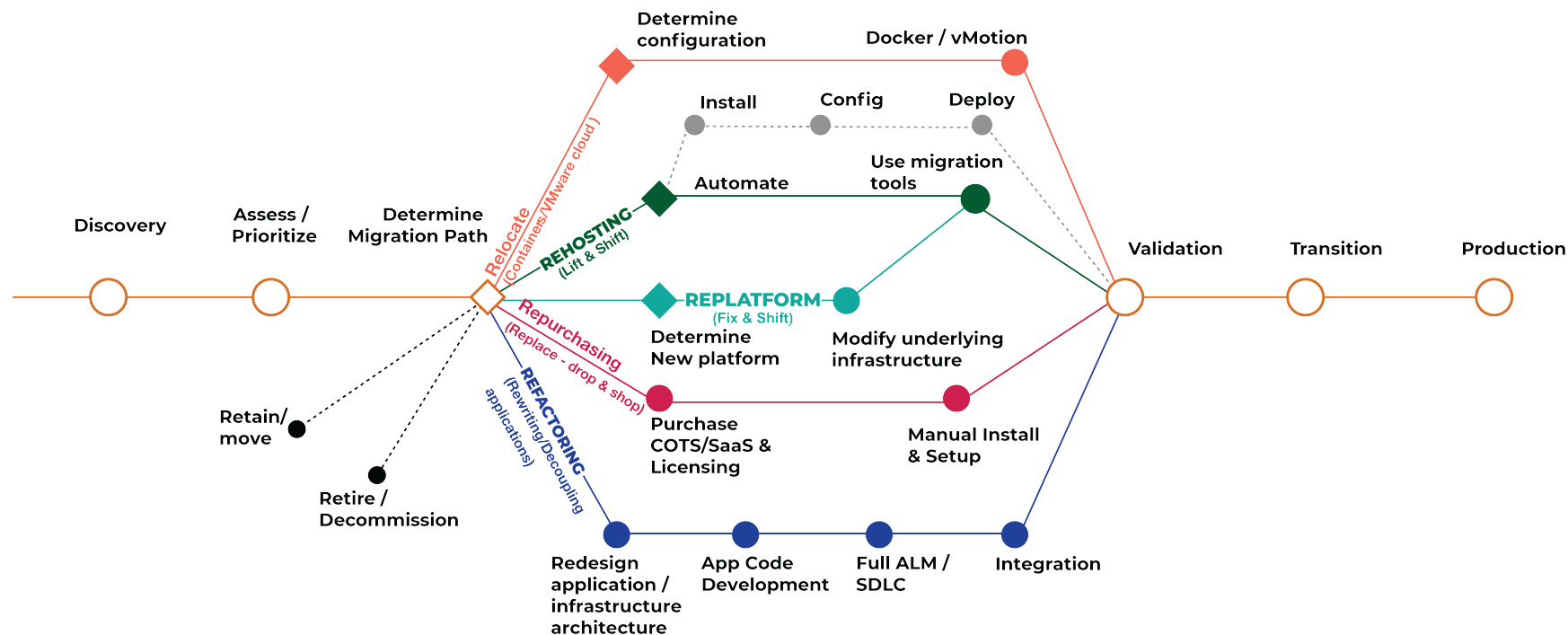
- ✓ Agility and Scalability
- ✓ Innovation & Emerging Technologies
- ✓ FedRAMP and FISMA ATO
Compliant Environment
- ✓ Interoperability and Portability

- ✓ Infrastructure as Code
- ✓ Security and Compliance (ATO)
Support
- ✓ Disaster Recovery
- ✓ High Availability
- ✓ CI/CD with Zero Downtime
Deployments

CLOUD ADOPTION JOURNEY

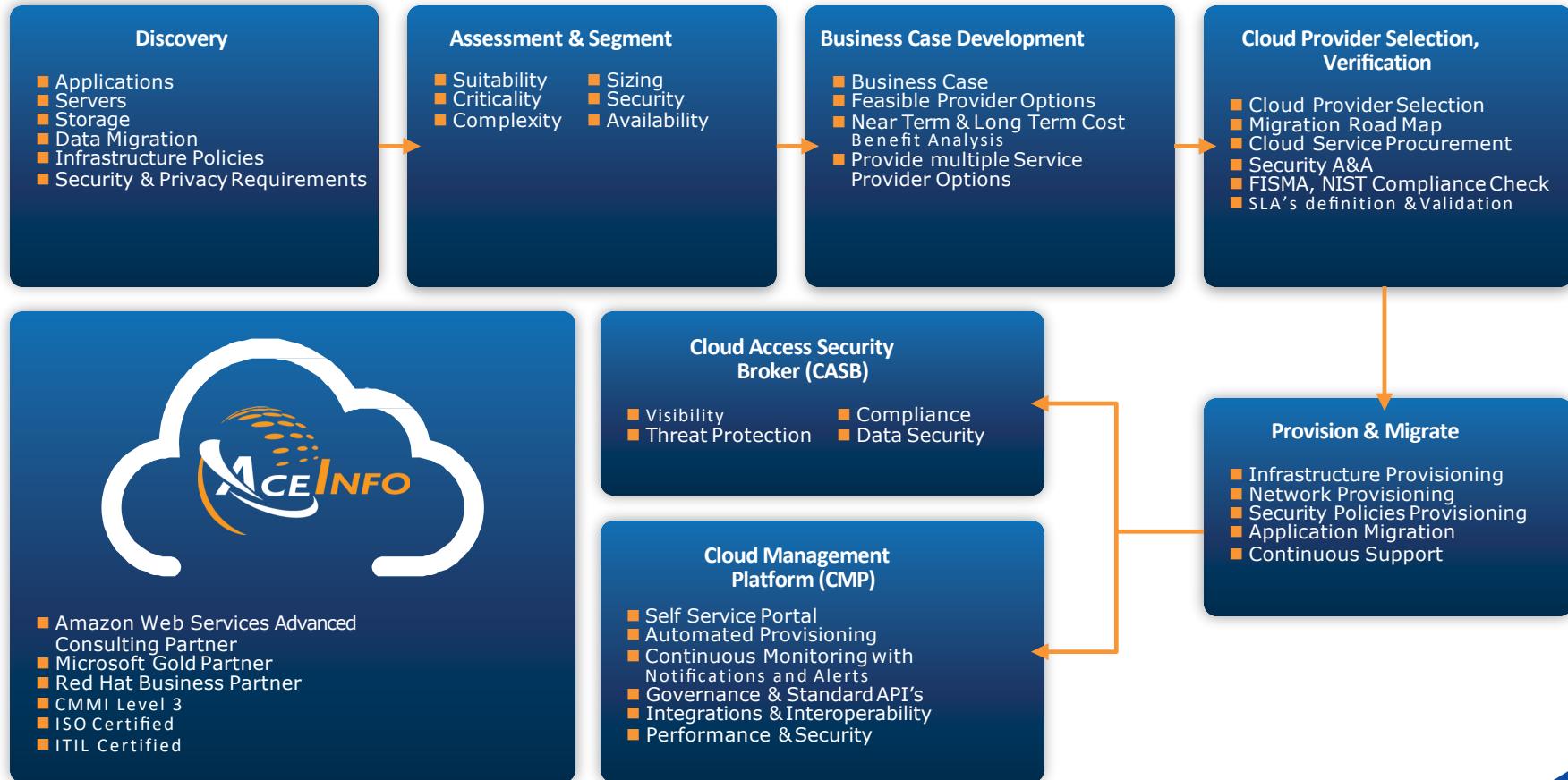


CLOUD ADOPTION STRATEGY (7 R's)



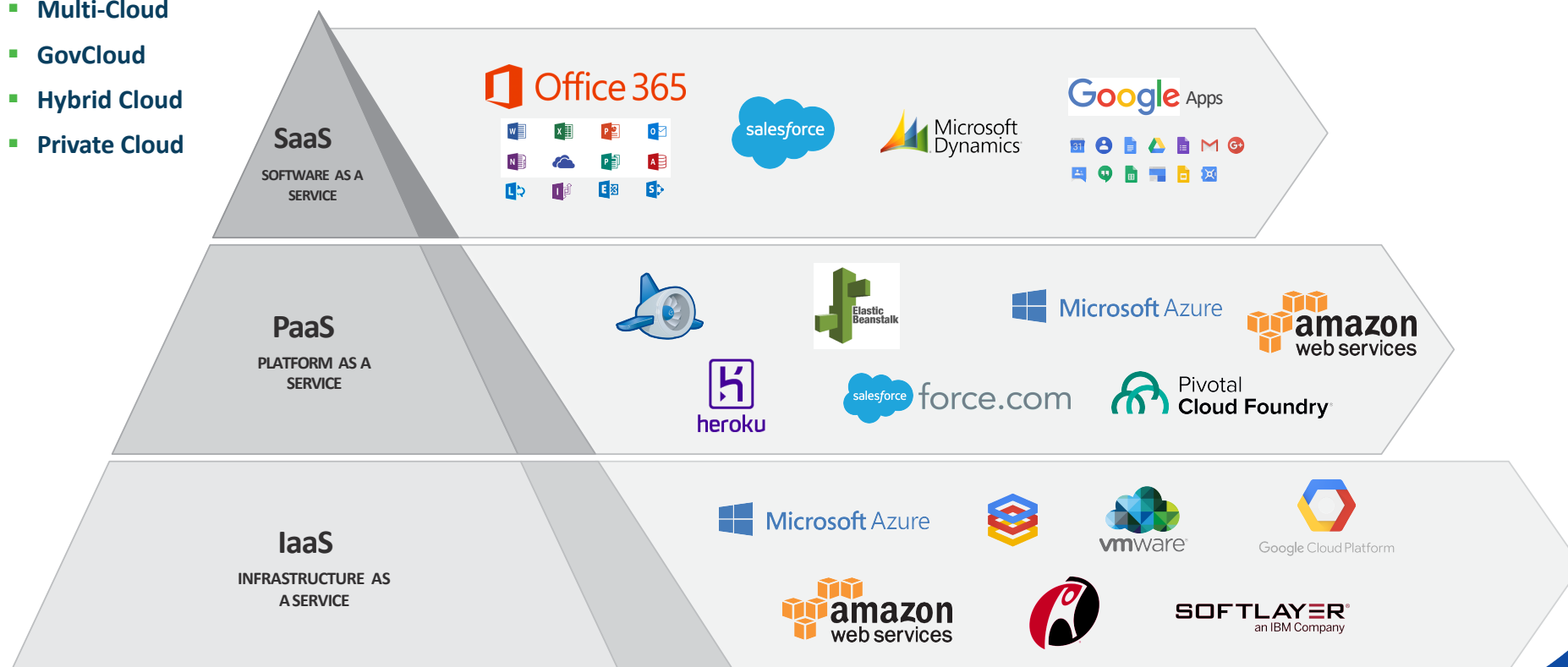
Migration Tools & Services: Storage Migration, Server Migration Services, Database Migration Service, Application Discovery Service, AWS Migration Hub, Azure Site Recovery and Cloud Endure

HIGH LEVEL CLOUD MIGRATION APPROACH



CLOUD SERVICE PROVIDERS, PLATFORMS AND TOOLS

- Multi-Cloud
- GovCloud
- Hybrid Cloud
- Private Cloud



CLOUD AND DEVSECOPS ADOPTION ACCELERATOR PLATFORM

<https://cloudaccelerator.aceinfosolutions.com>



Features and Benefits

Turnkey Solutions
Self-Service Templates
Empowers Dev, Ops and Sec
Accelerate Innovation
Rapid Development & Deployment

Container Orchestration

Red Hat OpenShift

DHS FEMA :350 - 500Containers
U.S. Air Force :100+ Containers
USDA :100+ Containers
CoE Lab :30 Containers

AWS ECS & EKS, GCP Kubernetes

CoE Lab :20 Containers

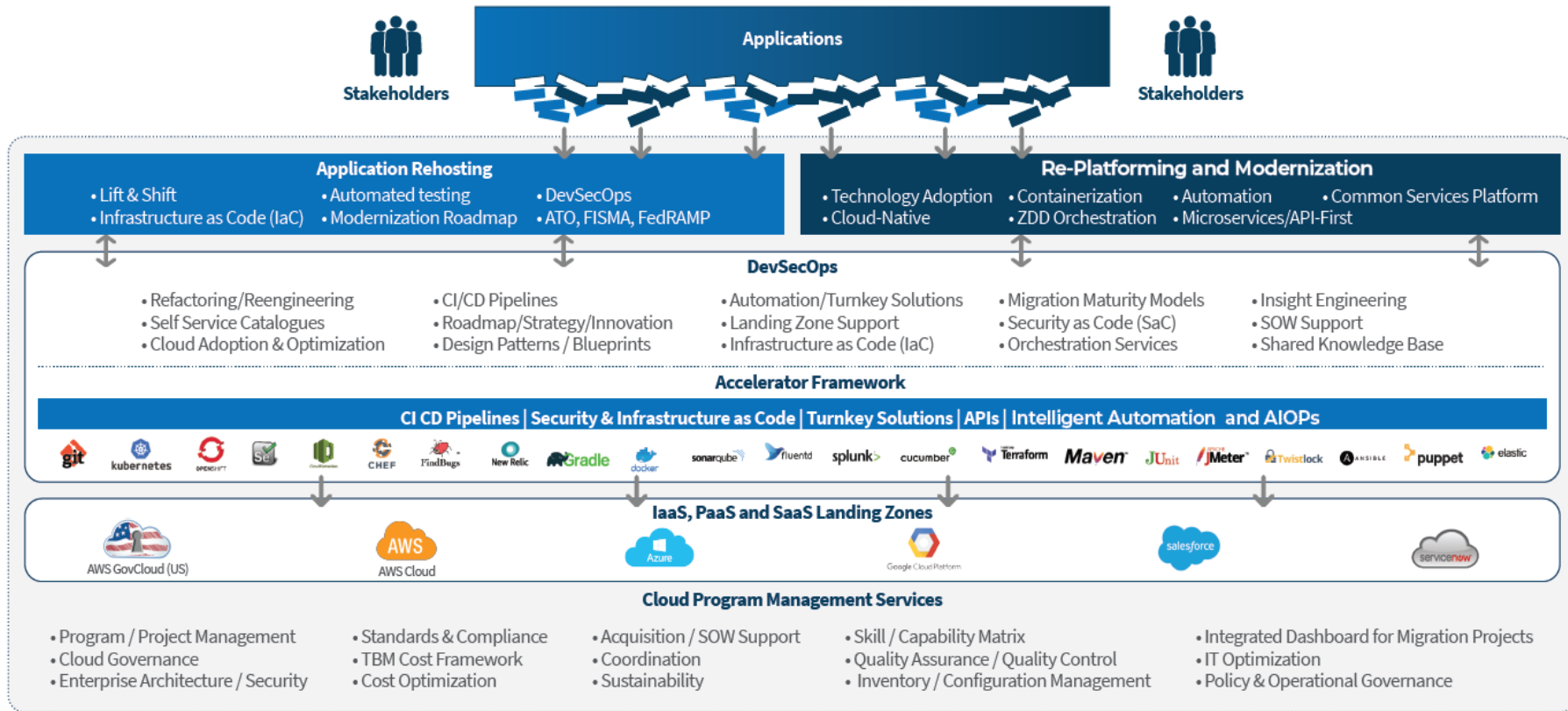
Cloud ACCELERATOR
Provided by AceInfo Solutions, a Dovel company

Cloud Adoption and Digital Modernization Accelerator
Accelerate Innovation | Empower Engineers | Simplify Cloud Journey
Concept Overview | Cloud Adoption Playbook

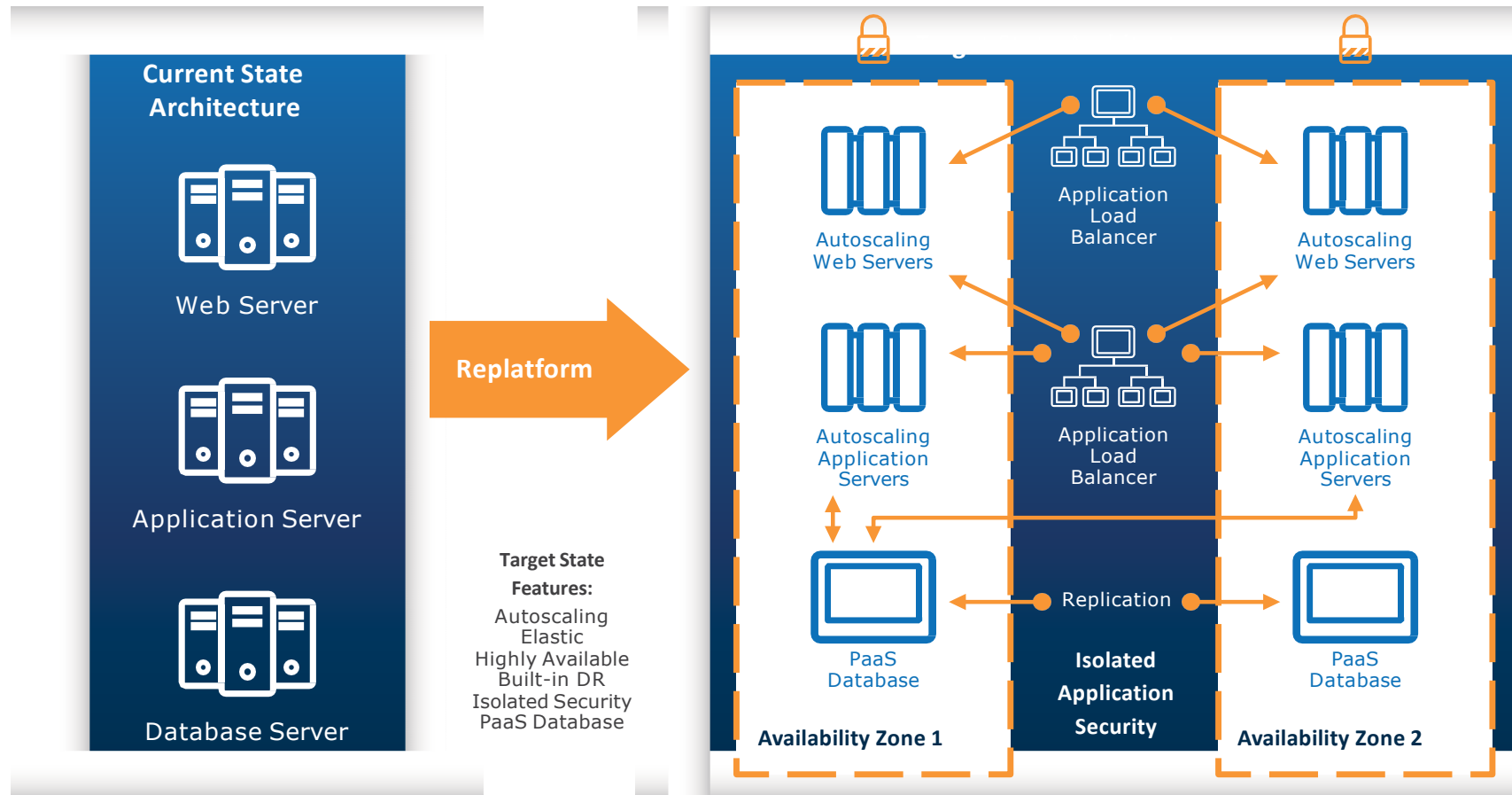
Home Resources vmanne@aceinfosolutions.com Logout

- Infrastructure as Code**
 - Create Cloud Infrastructure
 - View Cloud Infrastructure
 - Nexus/Docker Hub
 - Terraform Templates
 - Cloudformation Templates
 - Ansible Templates
 - Resources +
- Application Development**
 - Create Microservice and Application
 - Running Applications
 - Github Repository
 - Quality Assurance PAL
 - Discover Dovel Technologies
 - Application Prototypes +
 - Resources +
- DevSecOps**
 - Code Security Scan
 - View Code Scan Results
 - View All Pipelines
 - OpenShift
 - API Management
 - Regulatory Compliance
 - Resources +
- Monitoring and Logging**
 - Hygieia
 - Splunk
 - New Relic
 - ELK Stack
 - Slack
 - Sonarqube
 - Resources +
- Intelligent Services / Analytics**
(Powered by AWS and GCP Services)
 - Text to Speech Conversion
 - Machine Learning Models and APIs
- High Performance and Quantum Computing**
 - AWS Cloud HPC
 - Google Cloud HPC
 - Azure Cloud HPC

CLOUD ADOPTION ACCELERATOR CONOPS

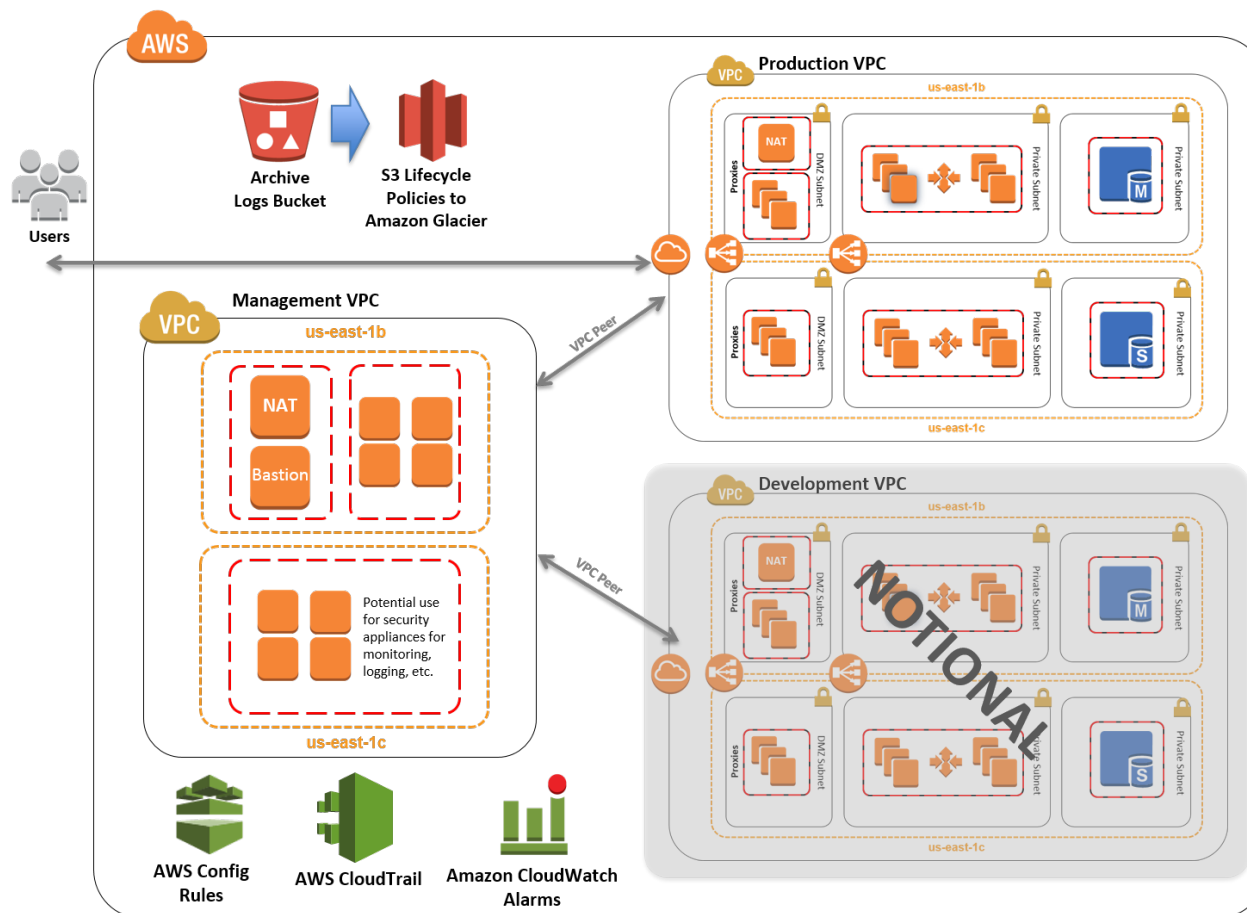


CLOUD PLATFORM TARGET ARCHITECTURE



STANDARDIZED ARCHITECTURE FOR NIST 800-53 FISMA LEVEL COMPLIANCE

BAKED-IN TEMPLATES FOR AUTHORITY TO OPERATE (ATO) ACCELERATION

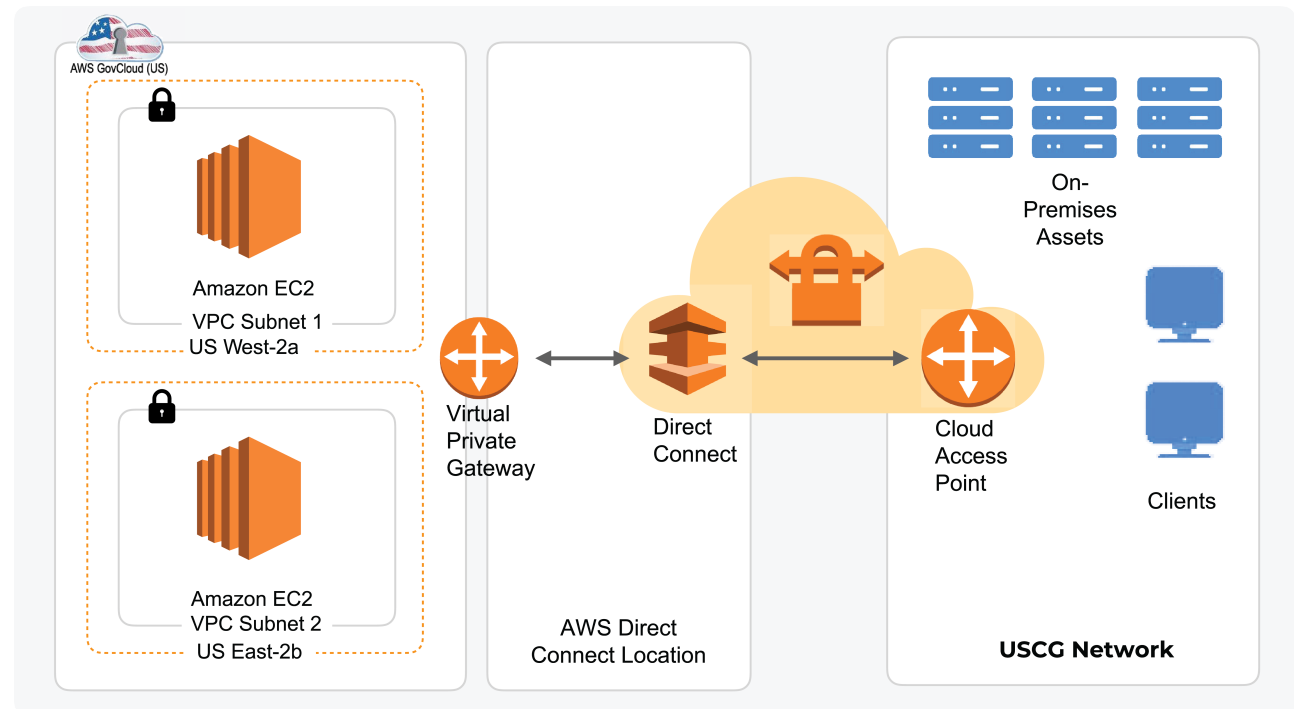


HYBRID CLOUD ARCHITECTURE

AWS Global Infrastructure

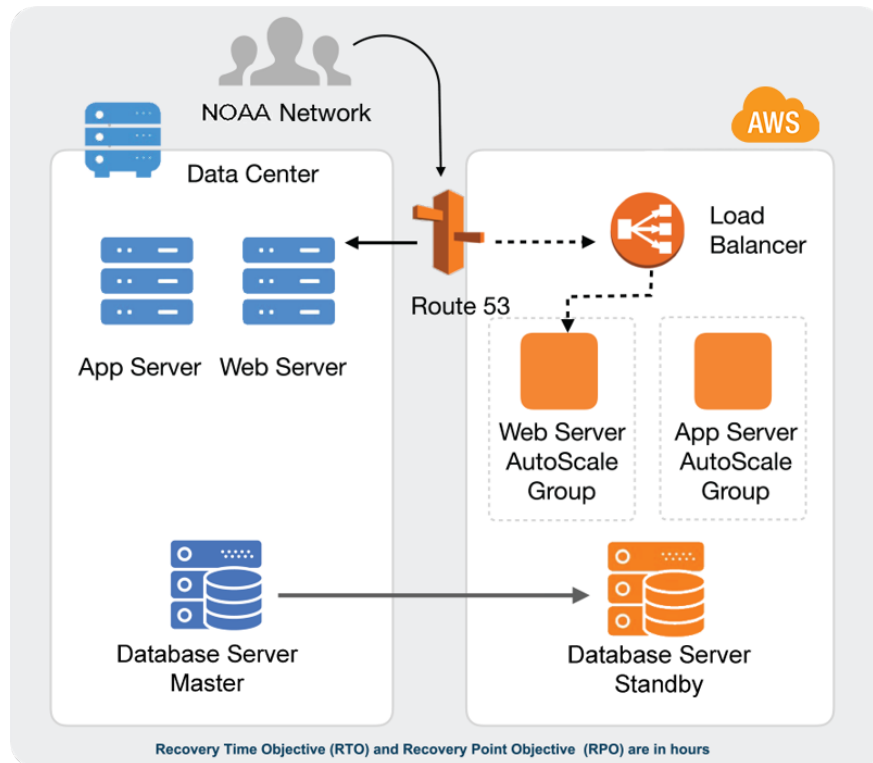
- 2 US GovCloud Regions (US East and West) with 6 Availability Zones
- 21 Geographic Regions with 66 Availability Zones.

- **Infrastructure as Code**
- **Security as Code**
- **Automation as Code**

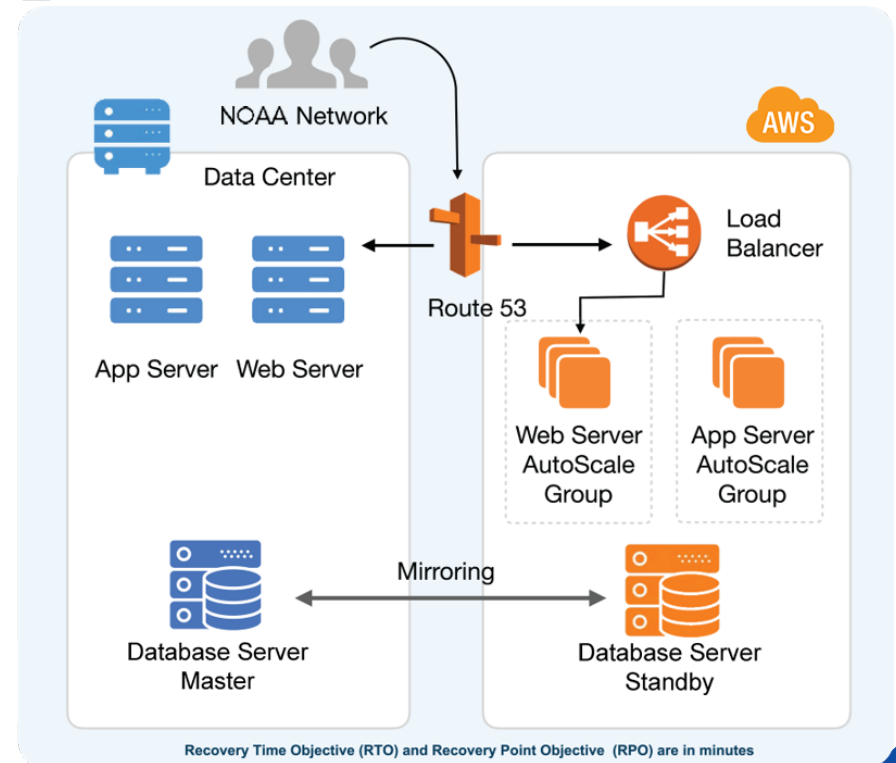


DISASTER RECOVERY AND FAILOVER ORCHESTRATION

Warm Standby



Active - Active



AGILE SOFTWARE DEVELOPMENT AND DEVSECOPS

15+ years of Agile Software Development

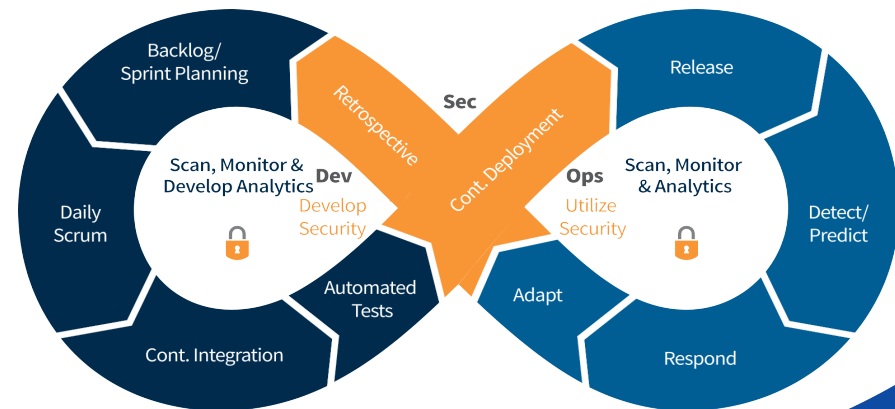
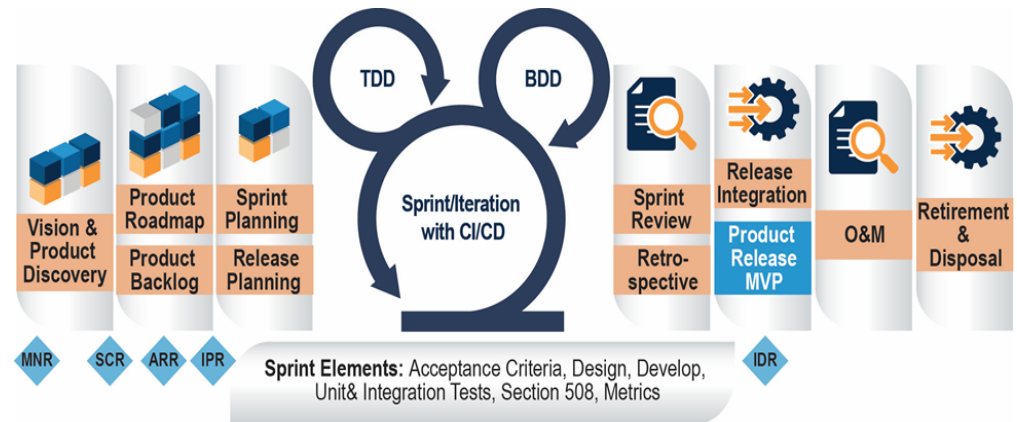
- Scrum, Kanban, Lean, SAFe
- Design Thinking
- User Centric Design (UX/UI)
- Responsive Web Design
- APIs, Microservices
- Opensource, COTS and GOTS
- Developed and maintain 500+ applications

350+ Agile Experienced and Certified Resources

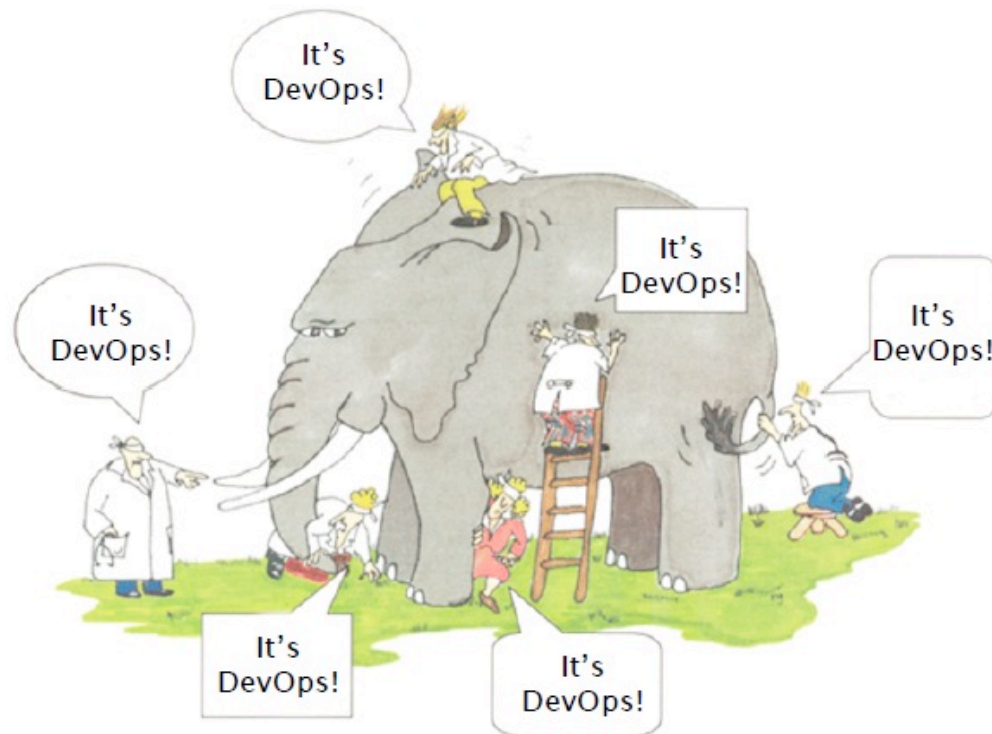
- Certified Scrum Masters/Professionals
- Agile Project Managers
- PMI-ACP and SAFe Certified Practitioners

5+ Years of DevOps / DevSecOps Implementation

- 30+ DevOps/DevSecOps Engineers
- 100+ Redhat Engineers (50+ Certified)
- 100+ Cloud Engineers (25+ Certified)



What is DevOps?



Let's try again - What is DevOps for customers ?

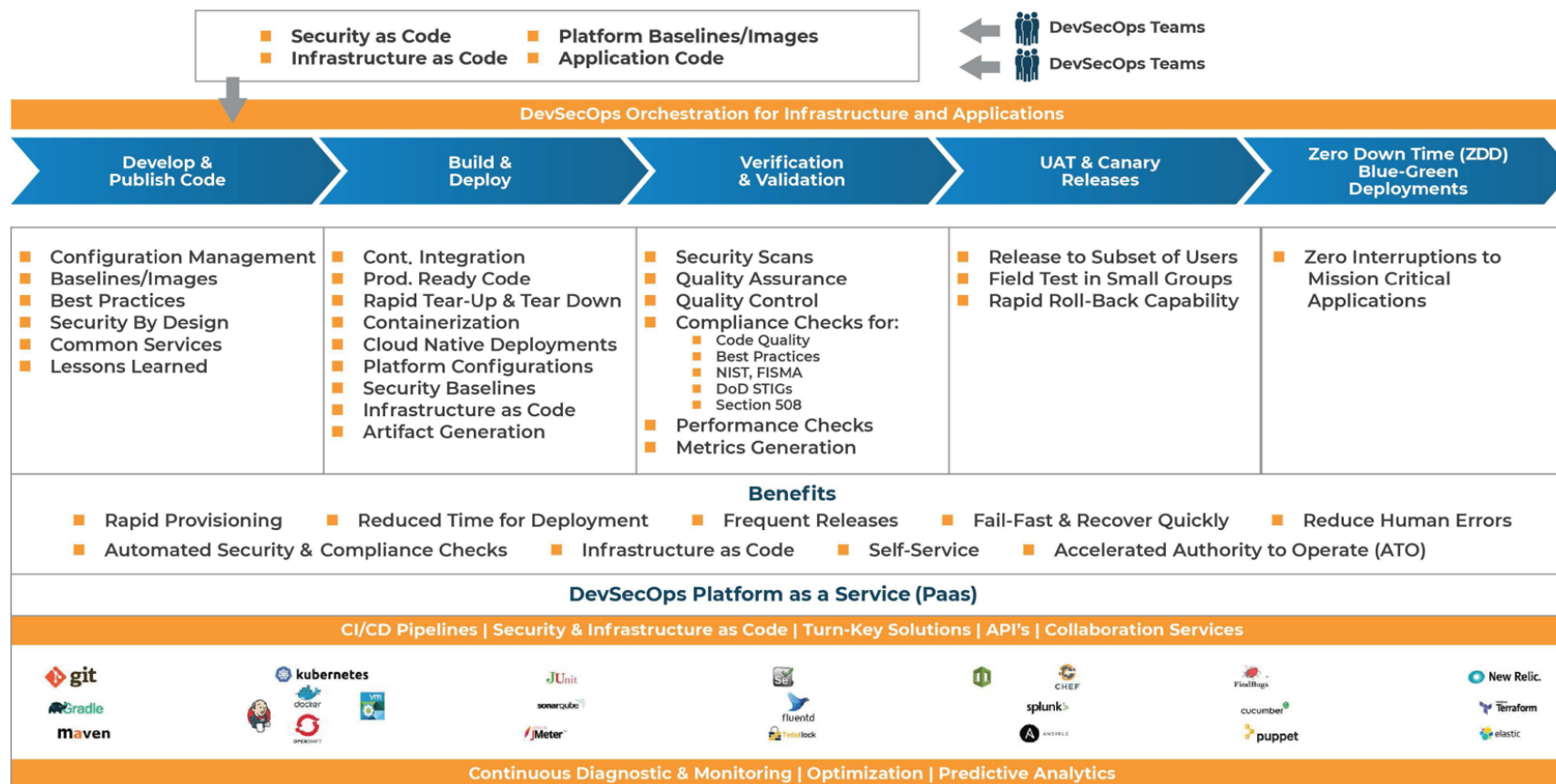
- ✓ Collaborative Culture
- ✓ Automation as Code
- ✓ Effective & Efficient Toolchain
- ✓ Monolithic to Microservices
- ✓ Containerization and ZDD
- ✓ High Availability in Cloud
- ✓ AIOps (Predict and Prevent)

Never send a human to do a machines job



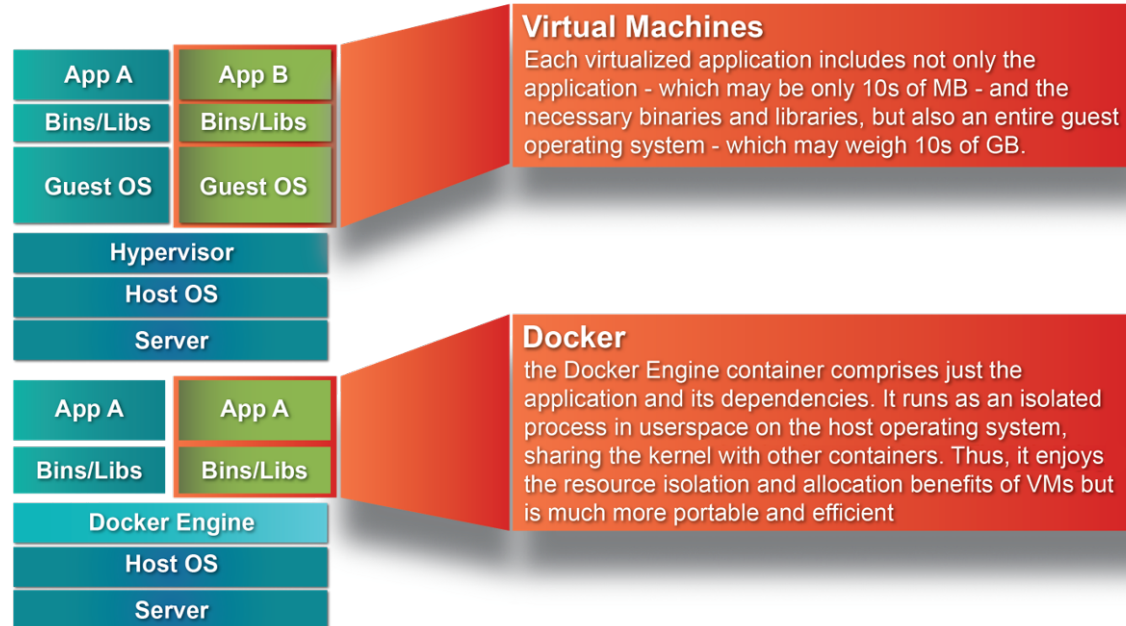
DEVSECOPS ORCHESTRATION FOR INFRASTRUCTURE & APPLICATIONS (CI/CD)

AGGREGATED ARTIFICATS FOR AUTHORITY TO OPERATE (ATO) ACCELERATION



CONTAINERIZE APPLICATION BUILDS AND SOFTWARE LIBRARIES

- Package Once and Deploy Anywhere
- Immutable and Reusable
- Standardized Environments
- Effective Configuration Management
- Portability and Integrity
- Cloud Readiness
- Interoperability
- Infrastructure as Code
- Security as Code / ATO
- Rapid Provisioning and Scaling



KUBERNETES BASED HIGH AVAILABILITY AND SELF-HEALING

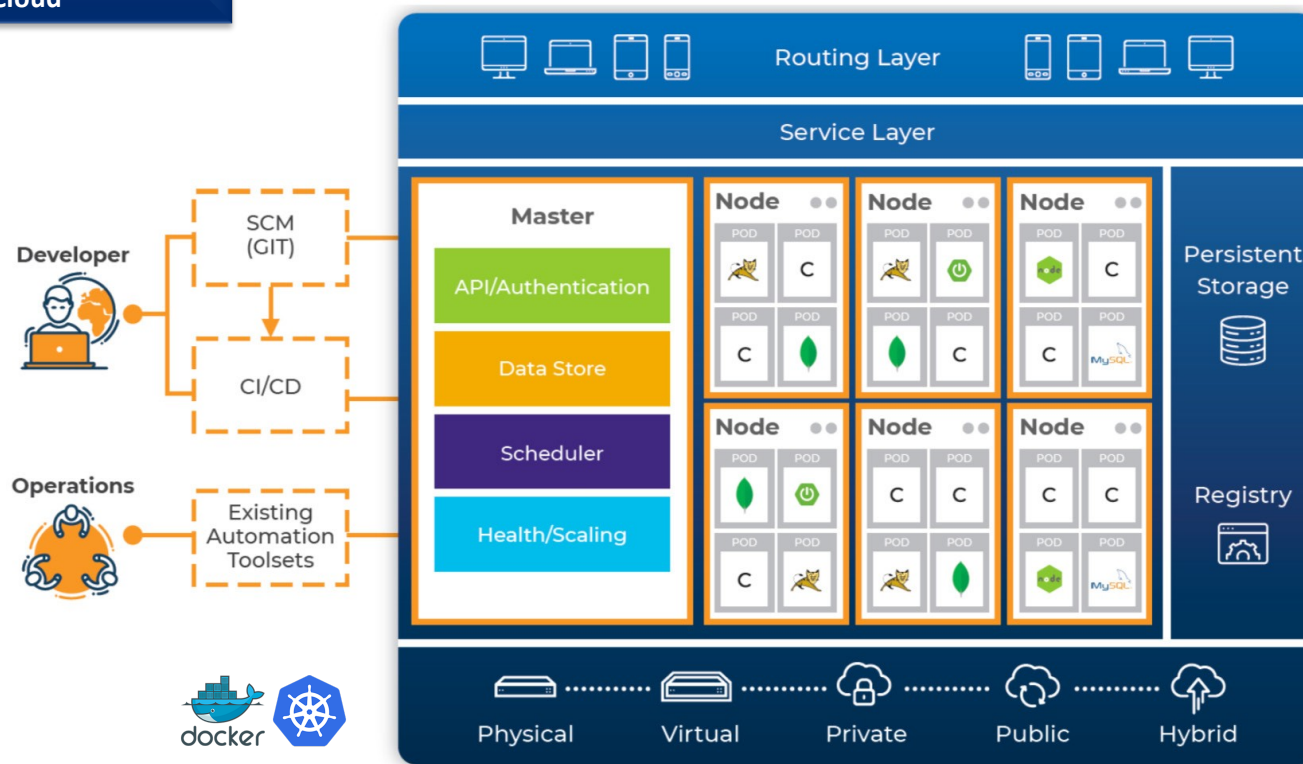
On Premise, Hybrid and Multi-Cloud

■ Turnkey Solutions

- Service Catalog / Templates
- Rapid Provisioning/Scaling
- Security and ATO
- Zero Downtime Deployments (ZDD)
- Self-Healing
- Monitoring and Logging
- AIOPs

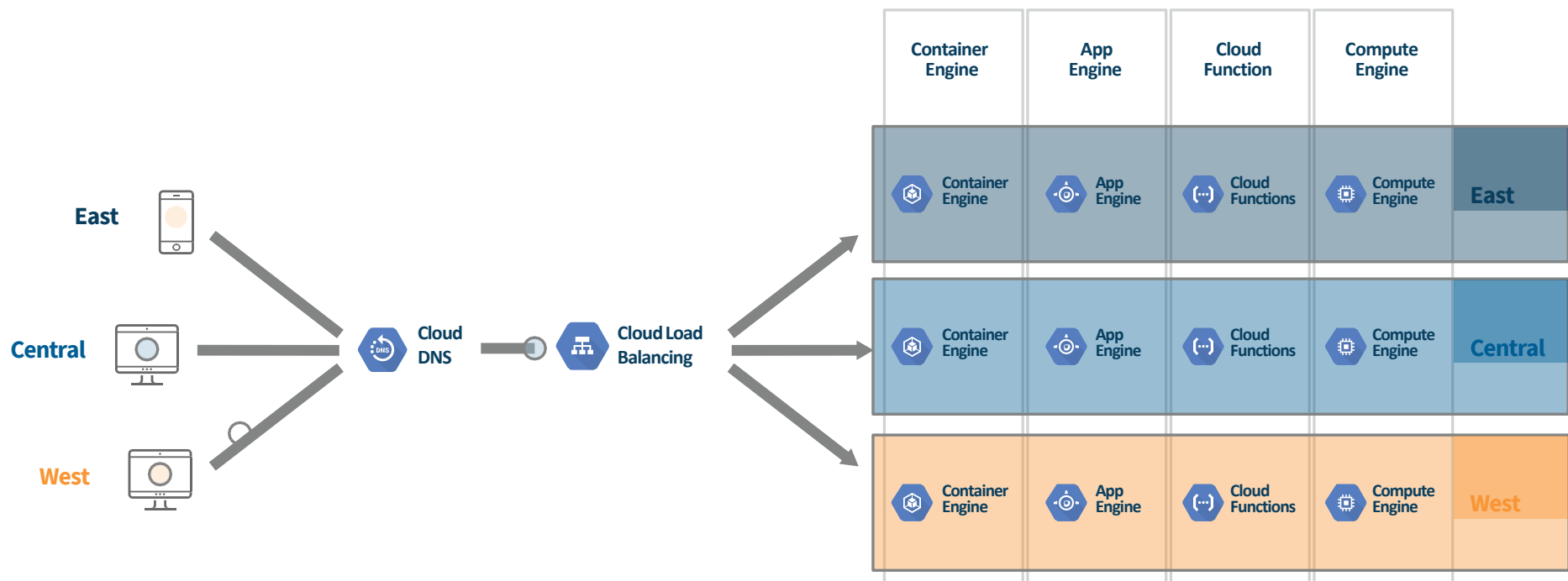
■ Turnkey Platforms

- Kubernetes
- Openshift
- Redhat Openshift
- Google Cloud (GKE)
- AWS (ECS, EKS)
- AZURE (CS, KS)

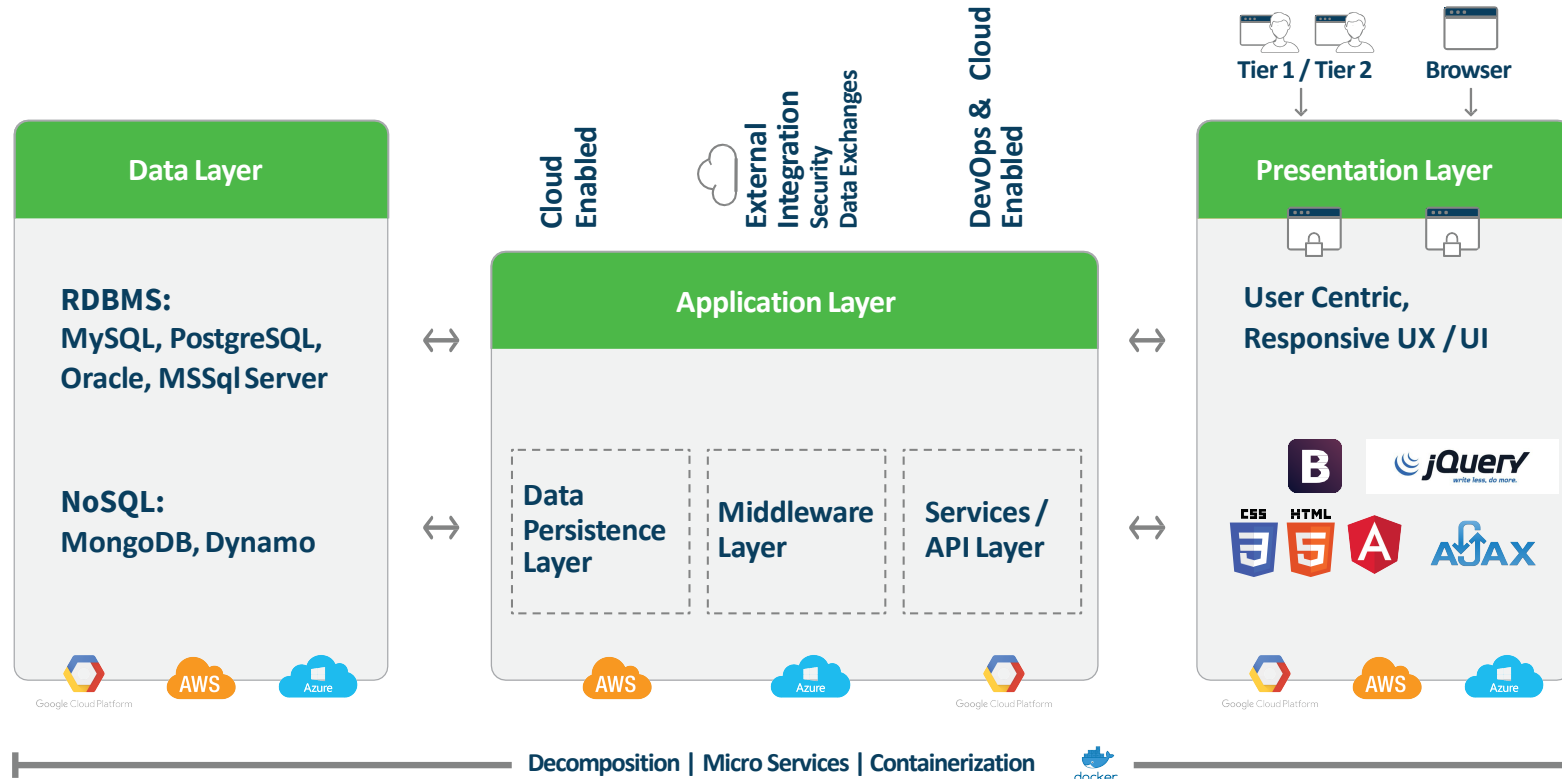


KUBERNETES CONTAINER ORCHESTRATION (GOOGLE CLOUD)

Multi Region Failover & High Availability



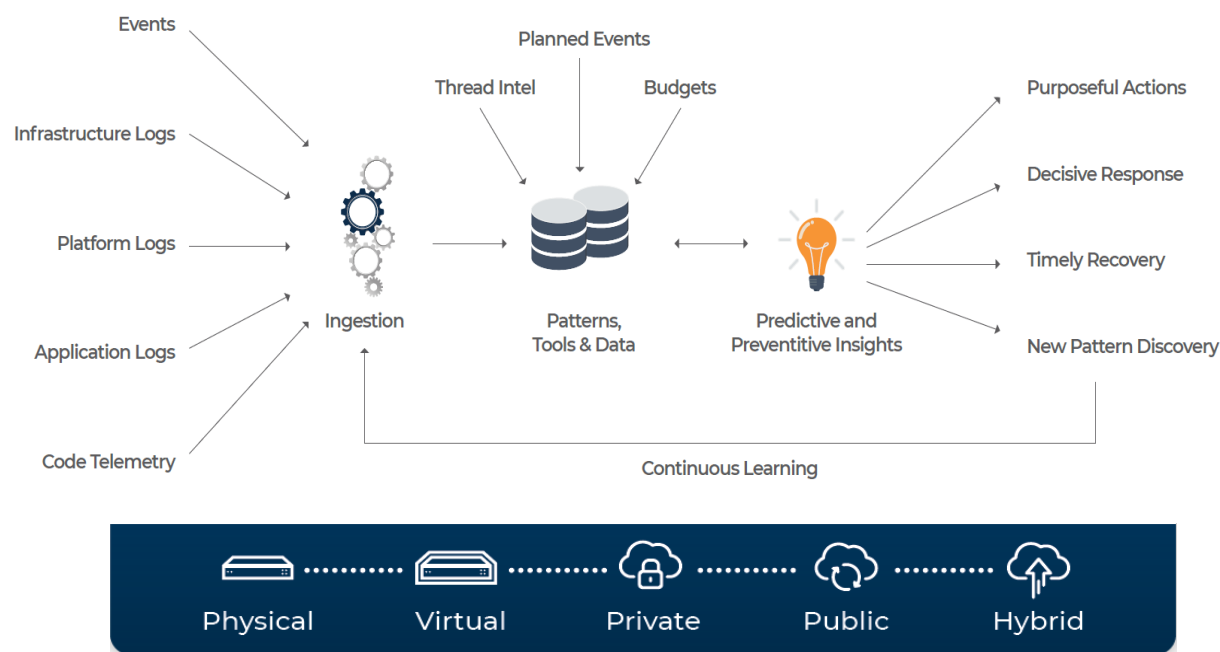
CLOUD NATIVE ARCHITECTURE FOR APPLICATION MODERNIZATION



ARTIFICIAL INTELLIGENCE FOR IT OPERATIONS (AIOPS)

PREDICTIVE INTELLIGENCE AND PREVENTIVE MEASURES

Full Stack Observability for Distributed Architectures



Use Data Lake and AI / ML

- ✓ Ingest & Process Data (3 Vs)
- ✓ Log Analytics and Insights
- ✓ Event Correlation
- ✓ Pattern Recognition
- ✓ Speed Root Cause Analysis
- ✓ Lower Noise | Detect Anomalies
- ✓ Automate Remediation Proactively
- ✓ Unify IT Metrics and Business KPIs
- ✓ Improve Customer Experience
- ✓ Reduce Cost | Improve Efficiency

CLOUD ADOPTION EXPERIENCE

Department of Homeland Security / Department of Defense

- DOD STIG Compliant Hybrid / Private Cloud, hosting 2500+ Servers, 75+ Applications for USCG
- Self-service provisioning of environments with Single Pane of Glass for Cloud Management
- DevSecOps: Docker, Kubernetes, OpenShift Container Orchestration
- Air Force CON-IT and FEMA PIVOT Systems modernized and implemented in GOGO cloud
- US Airforce, TSA, USCG, FEMA and DHS OIG

Department of Commerce

- 1500+ servers, 200+ Applications, Dev/Test Environments, 100+ Databases and petabytes of storage
- Data Lakes and Analytics
- Managed Services for AWS and Azure public and GovCloud
- Private, Public, & Hybrid Cloud
- DevSecOps: Ansible and OpenShift, Jenkins CICD
- Clients: Census and NOAA

USDA and DOJ COPS

- USDA Cloud Adoption BPA
- AWS Cloud for prototypes and evaluation of emerging technologies and innovation
- DevOps Automation, Jenkins CI/CD, Docker Containers and Microservices
- USDA NITC Cloud (GOGO) Migrations
- Migrated 40+ applications of DOJ COPS to Azure Cloud

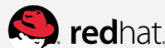
Department of Health and Human Services

- Consolidated and modernized legacy Grants Management systems for SAMHSA
- AWS GovCloud Implementation for SAMHSA
- Reduced TCO of existing systems by 40%
- Migrated legacy COMET application to AWS (<60 days) with zero service interruption



Advanced
Consulting
Partner

AWS Public Sector
Partner



READY
BUSINESS PARTNER

20+ Cloud Implementations | Private, Public and Hybrid Clouds
5+ Years for Cloud Computing, | AWS and MS Azure Consulting Partner
3000+ servers, 100+ applications, Petabyte storage in Cloud
100% AceInfo Websites/Applications in AWS/Azure
50 Cloud Skilled Engineers, 30 AWS/Azure/Google Certified Architects

Microsoft
Partner



Gold Application Development
Gold Cloud Platform
Gold Data Platform

CLOUD ADOPTION AND DIGITAL MODERNIZATION EXPERIENCE



AceInfo's Agile Experience	USDA EAS				USDA	USDA	NOAA	US EPA	DOJ	DOC	DHS	HHS
	DHS FEMA PIVOT	AirForce CON-IT	GSA	Other	FAS	FSA	NWS	OPP & OEI	COPS	US Census	US Coast Guard	SAMSHA
Number of Staff	60	60	33	80+	50+	60	50	25	40	124	140	45
Number of Agile Teams	6	6-7	4-5	30+	5	5	4	4	3	Varies	Varies	5
Agile SELC Methodologies	Scrum	Scrum	Scrum	Scrum, Kanban	Scrum	Scrum	Scrum, Kanban	Scrum, Kanban	Scrum	Hybrid	Hybrid	Scrum
DevOps/DevSecOps	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Portfolio Size	5 Applications	1 System	1 System	86 Applications	40+ Applications	140+ Applications	20 Applications	60+ Applications	50+ Applications	98 Applications	60 Applications	5 Systems
Degree of Complexity / Size	300+ User Stories 78 Sprints/year	300+ User Stories 120 Sprints/year	180+ User Stories 72 Sprints/year	3000+ User Stories 120 Sprints/year	1000+ User Stories 78 Sprints/year	1,800+ User Stories 120 Sprints/year	500+ User Stories 24 Sprints/year	800+ User Stories 27 Sprints/year	700+ User Stories 36 Sprints/year	Varies	Varies	1000+ User Stories 120 Sprints/year
Project Cloud Attributes												
SaaS		✓		✓				✓		✓		
PaaS	✓			✓				✓	✓	✓	✓	
IaaS	✓			✓				✓	✓	✓	✓	
COTS		✓	✓	✓	✓	✓		✓	✓	✓	✓	✓
Mobile			✓	✓					✓			
Custom	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓
DevOps	✓	✓		✓		✓	✓	✓	✓	✓	✓	✓
Support	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Public/Private Cloud	NITC Cloud	NITC Cloud	NITC Cloud	NITC Cloud	NITC Cloud	NITC Cloud	NOAA Cloud/ AWS	N/A	Azure GovCloud	AWS Cloud/ GovCloud	Hybrid: Private Cloud/AWS/Azure	AWS GovCloud
Number of Users	Nationwide	5,800	2,500	N/A	3,000	6,000	3 Million	1,300	50,000	20,000	150,000	20,000

CLOUD ADOPTION SERVICE MODELS WITH ESTIMATED PRICING

SINGLE APPLICATION MIGRATION



Service Description

- Strategy and Planning
- **Re-host (“Lift and shift”)** of Application including its database and integrations from on-premise to cloud
- **Re-platform** technology stack of on-premise application and migrate to cloud (Complexity: Moderate/High)

Factors Determining Migration Complexity (Low, Moderate,High)

- Number of Databases and Data Volume
- Number of Application Integrations (External & Internal), Web Pages, Userbase
- High Availability and Disaster Recovery requirements
- Re-hosting vs. Re-Platforming the application

Estimated Cost Per Application Per Annum (Labor Only) based on Complexity*

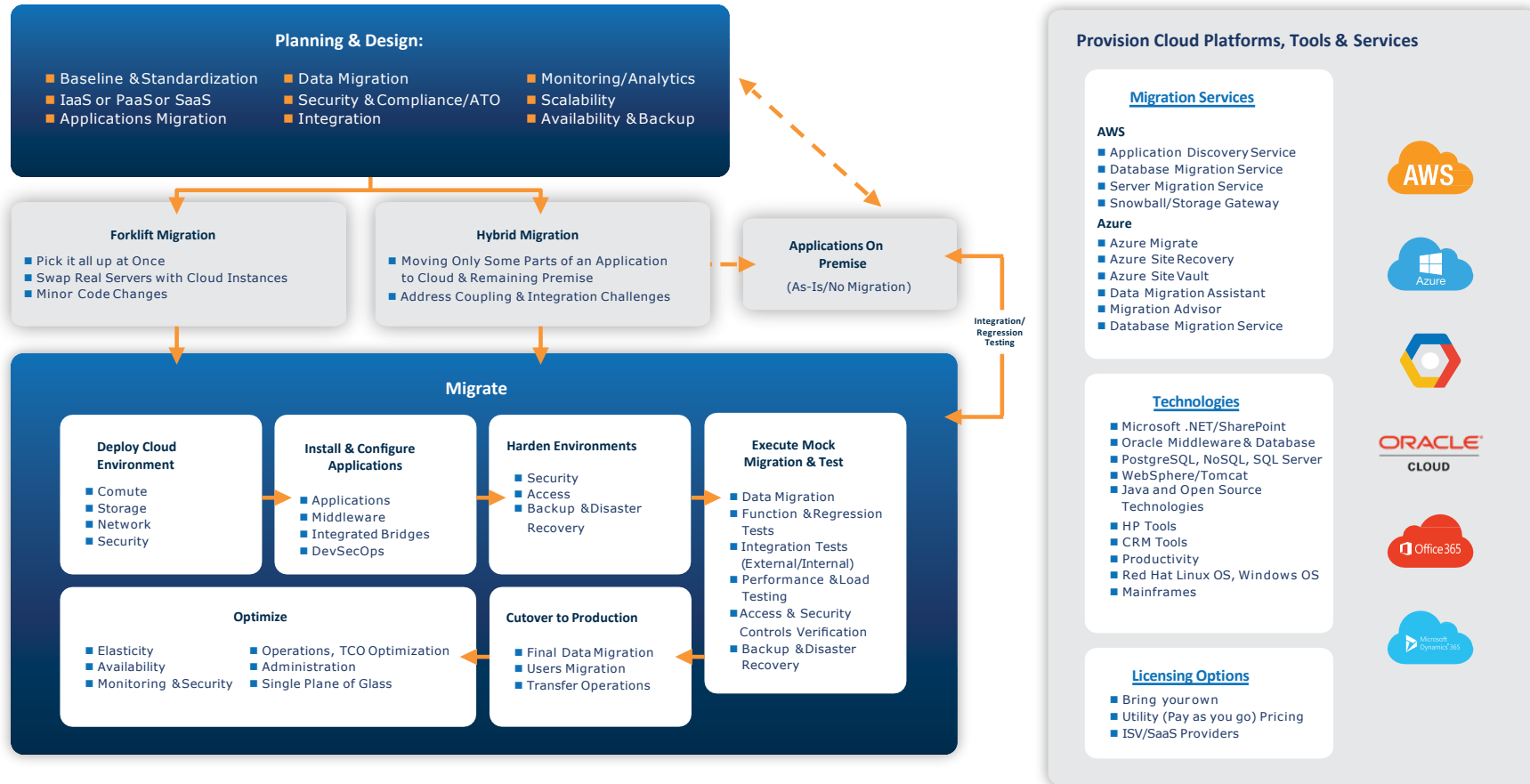
- Low: \$200K - \$400K
- Moderate: \$400K - \$800K
- High: \$800K and Above

Deliverables

- Production ready running App in Cloud (Public/GovCloud) including ATO
- Basic CI/CD Pipeline, High Availability and Disaster Recovery

*Cloud Infrastructure costs not included

RE-HOSTING (LIFT AND SHIFT) IMPLEMENTATION



APPLICATION MODERNIZATION



Service Description

- Strategy Consulting and Planning: Assist in conducting Discovery, Assessment and Recommend the Cloud Migration Path
- **Modernize** and **re-engineer** application using modern day and cloud enabled Architecture with APIs, Microservices and containers
- Deploy the modernized application in Cloud

Factors Determining Migration Complexity (Low, Moderate,High)

- Number of Databases and Data Volume
- Number of Application Integrations (External & Internal), Web Pages, Userbase
- High Availability and Disaster Recovery requirements
- Number of legacy technologies and frameworks being used

Estimated Cost Per Application (Labor Only) based on Complexity*

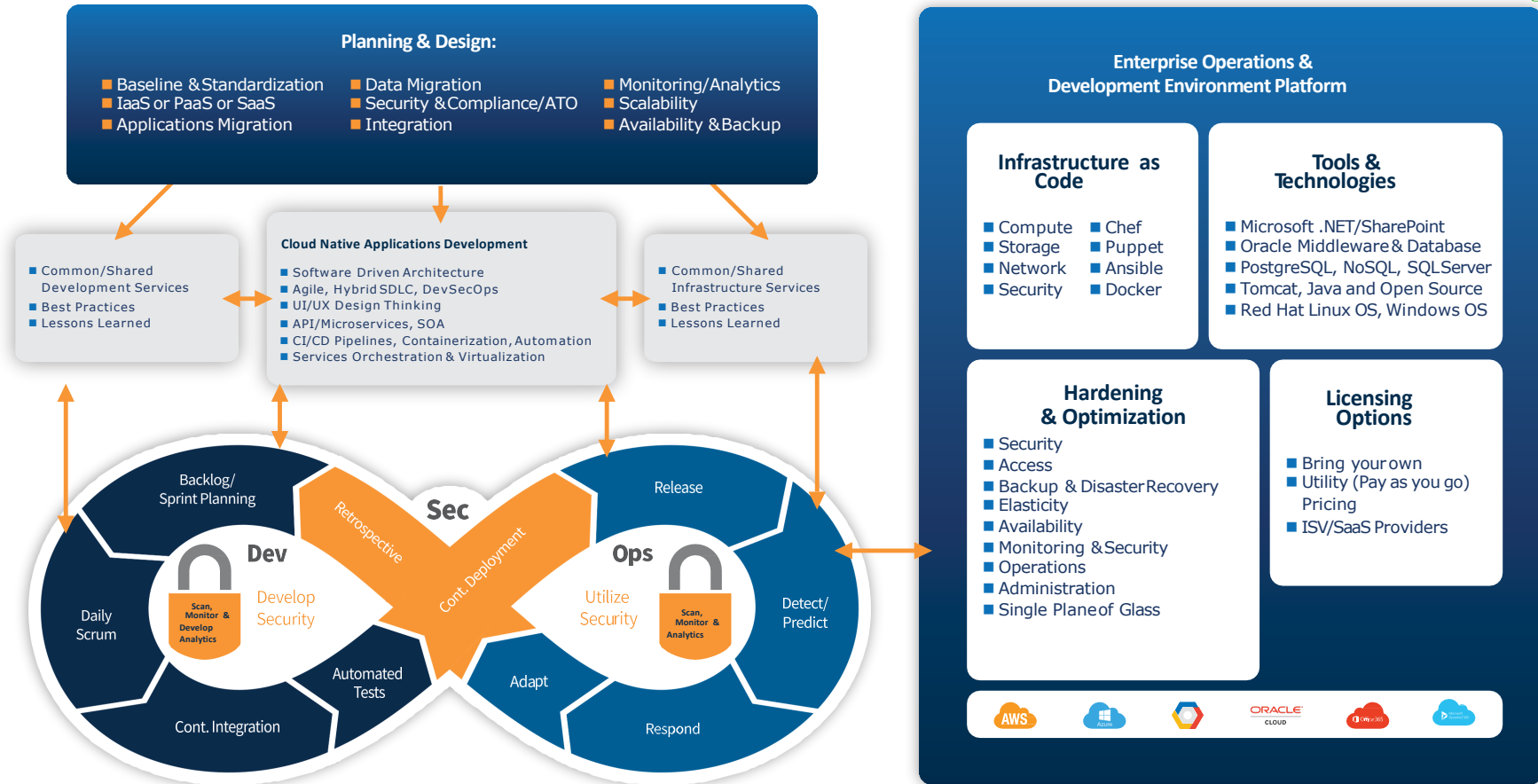
- Low: \$500K - \$800K
- Moderate: \$800K - \$2M
- High: \$2M and Above

Deliverables

- Production ready running Application in Cloud (Public/GovCloud) with **cloud native architecture** including **ATO**
- CI/CD Pipeline, High Availability and Disaster Recovery

*Cloud Infrastructure costs not included

RE-PLATFORM AND APPLICATION MODERNIZATION



FULL APPLICATION PORTFOLIO MIGRATION



Service Description

- Strategy Consulting & Planning including Discovery, Assessment/Prioritization and Determine Migration Path
- Application Migration (Lift & Shift), Re-Platform and Application Modernization
- **Deliver efficiencies** w.r.t Infrastructure as Code, Security as Code, Re-Usable processes and templates

Factors Determining Migration Complexity (Low, Moderate,High)

- Number of Applications in the portfolio, requiring Re-hosting, Re-Platforming and Modernize
- Number of Databases and Data Volume, Application Integrations (External & Internal), Web Pages, Userbase
- High Availability and Disaster Recovery requirements

Estimated Cost Per Application (Labor Only) based on Complexity*

Re-host / Re-Platform

- Low: \$200K - \$400K
- Moderate: \$400K - \$800K
- High: \$800K and Above

Modernize

- Low: \$500K - \$800K
- Moderate: \$800K - \$2M
- High: \$2M and Above

Deliverables

- Production ready Applications in Cloud (Public/GovCloud) with cloud native architecture including ATO
- **Single GSS** with applicable **ATO** for the entire portfolio
- **CI/CD Pipeline**, High Availability and Disaster Recovery

*Cloud Infrastructure costs not included

APPLICATIONS MAINTENANCE & SUPPORT IN CLOUD



Service Description

- Infrastructure and Platform support for applications in cloud including ATO, patching and monitoring
- Does not include application development or end user support

Factors Determining Migration Complexity (Low, Moderate,High)

- Number of server nodes for application and databases
- Number of integrations (external/internal)

Estimated Cost Per Application Per Annum (Labor Only) based on Complexity*

- Low: \$200K - \$300K
- Moderate: \$300K - \$500K
- High: \$500K and Above

Deliverables

- System Availability and Scalability
- Systems Administration including Backup & Recovery
- Continued ATO with applicable CDM (continuous diagnostics and monitoring)

*Cloud Infrastructure costs not included

CUSTOMER VALUE PROPOSITION

- Promote collaboration and innovation across the organization for technical excellence, customer success and business growth
- Incubate and operationalize solutions using emerging architectures and technologies and build intellectual capital for differentiative capabilities and customer success
- Simplify and accelerate Digital Modernization and complex Systems Integration through software driven architecture, reusable frameworks and turnkey solutions
- Leverage Cloud Computing, Automation and Intelligent Platforms to build solutions that learn, evolve, optimize and self-heal for increased efficiency and resilience
- Lead partnerships with industry, academia and research institutes to bring new ideas and experiment with them in our Emerging Technologies Lab for release